

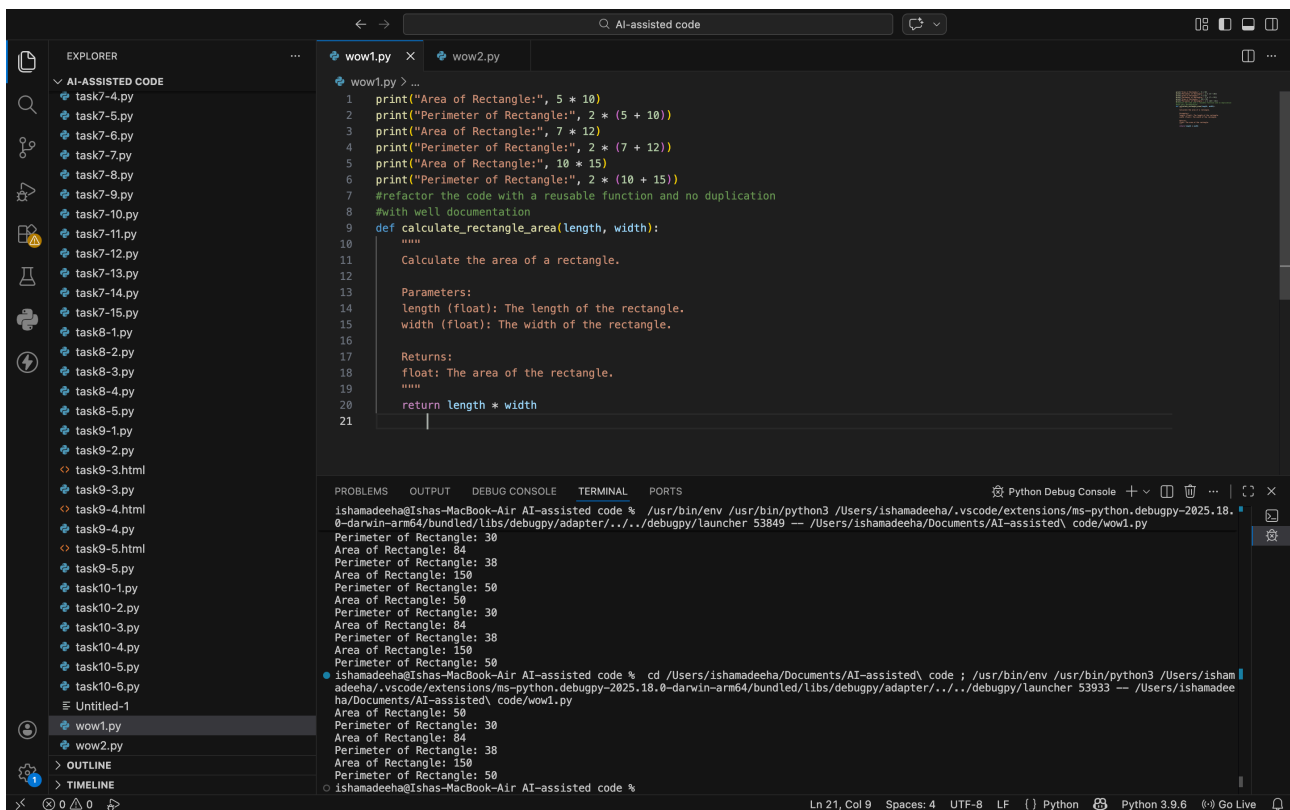
AIAC ASSIGNMENT - 13

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BATCH-01

TASK-1:



```
1 print("Area of Rectangle:", 5 * 10)
2 print("Perimeter of Rectangle:", 2 * (5 + 10))
3 print("Area of Rectangle:", 7 * 12)
4 print("Perimeter of Rectangle:", 2 * (7 + 12))
5 print("Area of Rectangle:", 10 * 15)
6 print("Perimeter of Rectangle:", 2 * (10 + 15))
7 #refactor the code with a reusable function and no duplication
8 #with well documentation
9 def calculate_rectangle_area(length, width):
10     """
11     Calculate the area of a rectangle.
12
13     Parameters:
14     length (float): The length of the rectangle.
15     width (float): The width of the rectangle.
16
17     Returns:
18     float: The area of the rectangle.
19     """
20     return length * width
21
```

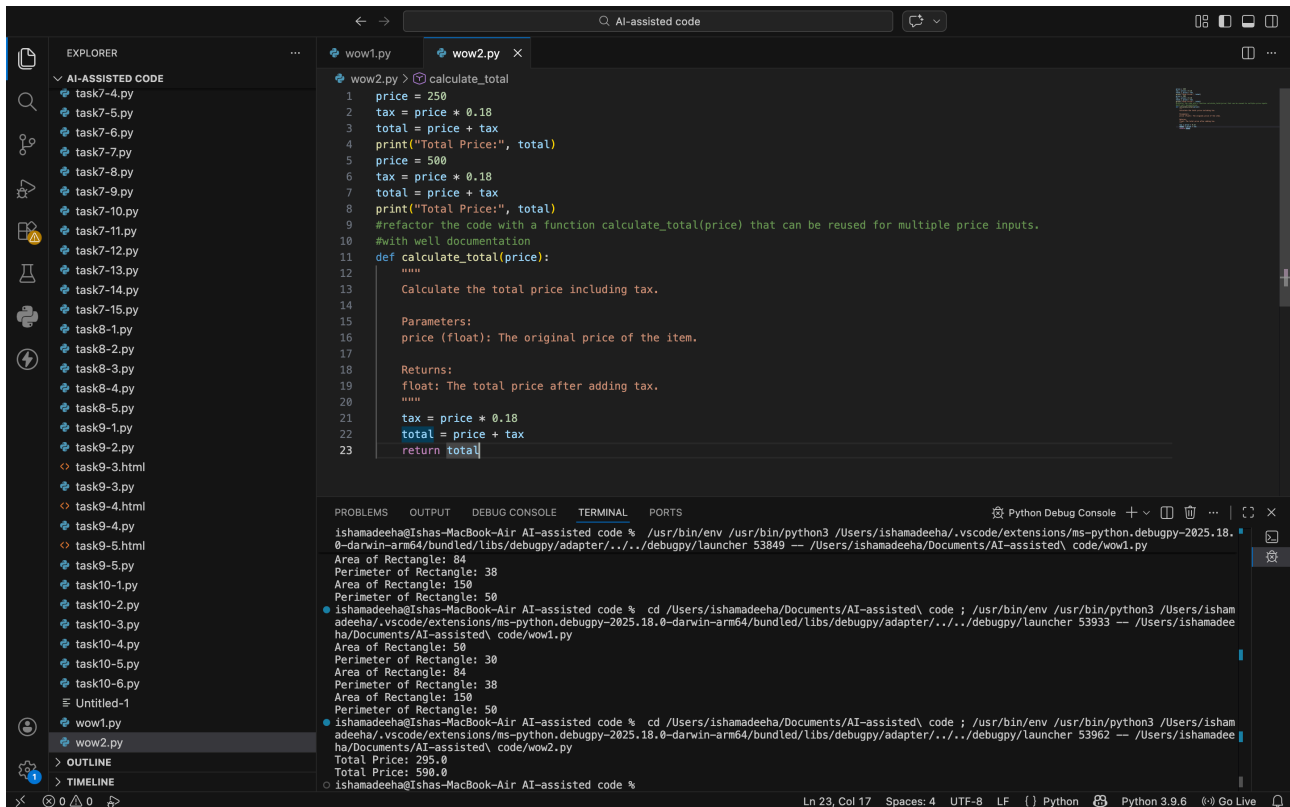
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python Debug Console

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 53849 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow1.py
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 53933 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow1.py
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

Ln 21, Col 9 Spaces: 4 UTF-8 LF Python Python 3.9.6 Go Live

TASK-02:



```
1 price = 250
2 tax = price * 0.18
3 total = price + tax
4 print("Total Price:", total)
5 price = 500
6 tax = price * 0.18
7 total = price + tax
8 print("Total Price:", total)
9 #refactor the code with a function calculate_total(price) that can be reused for multiple price inputs.
10 #with well documentation
11 def calculate_total(price):
12     """
13     Calculate the total price including tax.
14
15     Parameters:
16     price (float): The original price of the item.
17
18     Returns:
19     float: The total price after adding tax.
20     """
21     tax = price * 0.18
22     total = price + tax
23     return total
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 53849 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow1.py

Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 53933 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow1.py

Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50

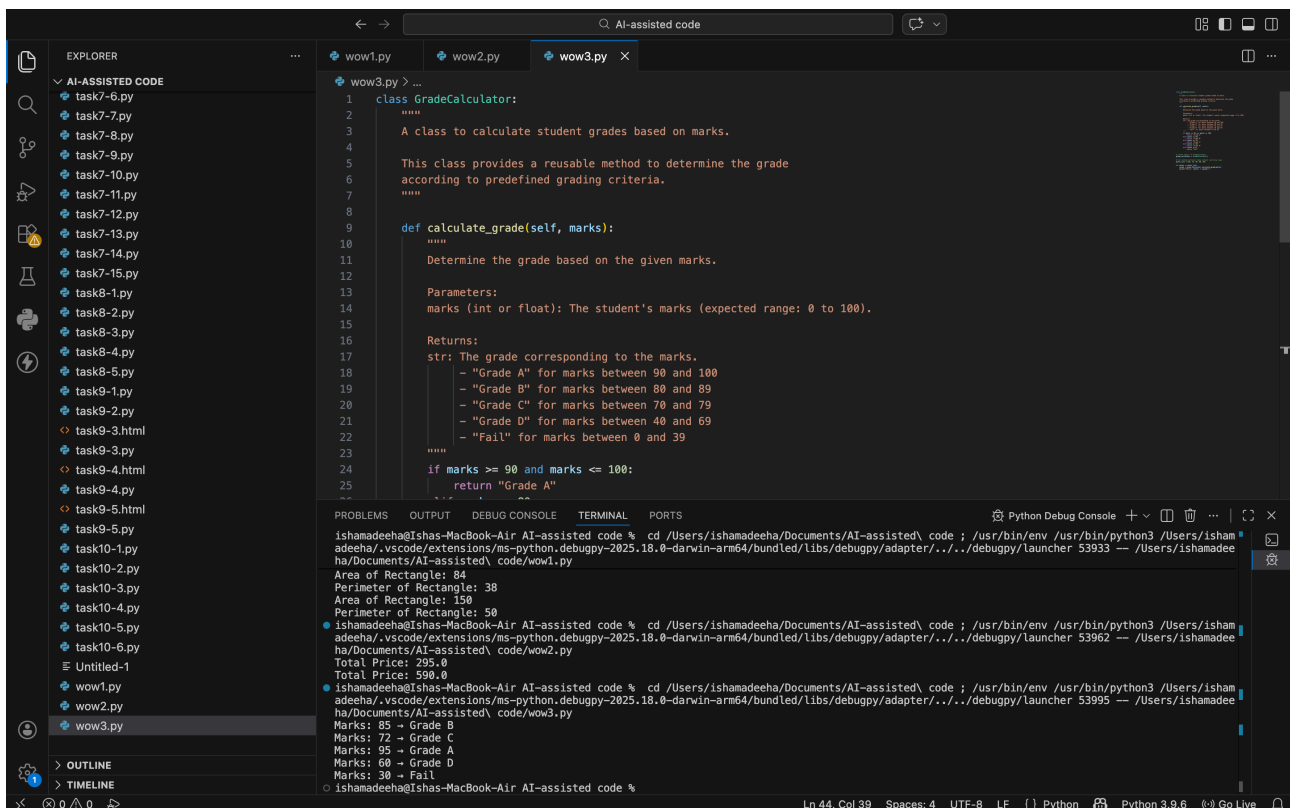
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 53962 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow2.py

Total Price: 295.0
Total Price: 590.0

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 23, Col 17 Spaces: 4 UTF-8 LF Python Python 3.9.6 Go Live

TASK-03:



```
1 class GradeCalculator:
2     """
3     A class to calculate student grades based on marks.
4
5     This class provides a reusable method to determine the grade
6     according to predefined grading criteria.
7     """
8
9     def calculate_grade(self, marks):
10         """
11         Determine the grade based on the given marks.
12
13         Parameters:
14         marks (int or float): The student's marks (expected range: 0 to 100).
15
16         Returns:
17         str: The grade corresponding to the marks.
18             - "Grade A" for marks between 90 and 100
19             - "Grade B" for marks between 80 and 89
20             - "Grade C" for marks between 70 and 79
21             - "Grade D" for marks between 60 and 69
22             - "Fail" for marks between 0 and 59
23         """
24         if marks >= 90 and marks <= 100:
25             return "Grade A"
26         elif marks >= 80 and marks <= 89:
27             return "Grade B"
28         elif marks >= 70 and marks <= 79:
29             return "Grade C"
30         elif marks >= 60 and marks <= 69:
31             return "Grade D"
32         else:
33             return "Fail"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 53933 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow1.py

Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 53962 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow2.py

Total Price: 295.0
Total Price: 590.0

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 53995 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow3.py

Marks: 85 -> Grade B
Marks: 72 -> Grade C
Marks: 95 -> Grade A
Marks: 60 -> Grade D
Marks: 30 -> Fail

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 44, Col 39 Spaces: 4 UTF-8 LF Python Python 3.9.6 Go Live

TASK-04:

The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files task7-6.py through task10-6.py, and task9-3.html through task9-5.html. The main editor window displays 'wow4.py' with the following code:

```
wow4.py > get_input
1 num = int(input("Enter number: "))
2 square = num * num
3 print("Square:", square)
4 #refactor the code using functions like get_input(), calculate_square(), and display_result()
5 #Identify input, processing, and output sections
6 #Convert each into a separate function
7 #With proper documentation
8 def get_input():
9     """
10    Get a number input from the user.
11
12    Returns:
13    int: The number entered by the user.
14    """
15    return int(input("Enter number: "))
16 def calculate_square(num):
17     """Calculate the square of a number.
18
19    Parameters:
20    num (int): The number to square.
21
22    Returns:
23    int: The square of the number.
24    """
25    return num * num
```

The bottom panel shows the 'TERMINAL' output:

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 53962 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow2.py
Total Price: 295.0
Total Price: 590.0
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 53995 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow2.py
Marks: 85 -> Grade B
Marks: 72 -> Grade C
Marks: 95 -> Grade A
Marks: 60 -> Grade D
Marks: 30 -> Fail
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 54109 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow4.py
Enter number: 25
Square: 625
Enter number: 3
Square: 9
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-05:

The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'AI-ASSISTED CODE' with files task7-8.py through task10-6.py, and task9-3.html through task9-5.html. The main editor window displays 'wow5.py' with the following code:

```
wow5.py > ...
1 salary = 50000
2 tax = salary * 0.2
3 net = salary - tax
4 print(net)
5 #refactor the code using a class like EmployeeSalaryCalculator with methods and attributes(encapsulation) to calculate net
6 #With proper documentation
7 class EmployeeSalaryCalculator:
8     """
9     A class to calculate the net salary of an employee after tax deduction.
10
11    This class encapsulates the salary and tax calculation logic, allowing for
12    reusable methods to compute the net salary based on a given gross salary.
13    """
14
15    def __init__(self, gross_salary):
16        """
17        Initialize the EmployeeSalaryCalculator with the gross salary.
18
19        Parameters:
20        gross_salary (float): The gross salary of the employee.
21        """
22        self.gross_salary = gross_salary
23
24    def calculate_tax(self):
25        """
```

The bottom panel shows the 'TERMINAL' output:

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 53995 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow3.py
Marks: 85 -> Grade B
Marks: 72 -> Grade C
Marks: 95 -> Grade A
Marks: 60 -> Grade D
Marks: 30 -> Fail
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 54109 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow4.py
Enter number: 25
Square: 625
Enter number: 3
Square: 9
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapter/.../debugpy/launcher 54218 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow5.py
40000.0
40000.0
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-06:

```
1 # library.py
2
3 library_db = {}
4
5
6 def add_book(title, author, isbn):
7     """
8     Adds a book to the library database.
9     Parameters:
10         title (str): The title of the book.
11         author (str): The author of the book.
12         isbn (str): The unique ISBN number of the book.
13     Returns:
14         str: A message indicating whether the book was added successfully or if it already exists.
15     """
16     if isbn not in library_db:
17         library_db[isbn] = {"title": title, "author": author}
18         return "Book added successfully."
19     else:
20         return "Book already exists."
21
22
23 def remove_book(isbn):
24     """
25     Removes a book from the library database based on its ISBN.
26     Parameters:
27         isbn (str): The unique ISBN number of the book to be removed.
28     Returns:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python Debug Console

- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54604 -- /Users/ishamadeeha/Documents/AI-assisted\ code/library.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 main.py
- Book added successfully.
- Book added successfully.
- {'title': 'Python Basics', 'author': 'John Doe'}
- Book removed successfully.
- Book not found.
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pydoc -w library
- wrote library.html
- ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 50, Col 33 Spaces: 4 UTF-8 LF {} Python Python 3.9.6 Go Live

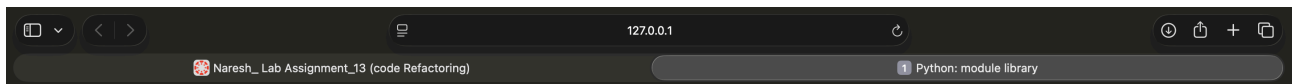
TASK-07:

```
1 # library.py
2
3 library_db = {}
4
5
6 def add_book(title, author, isbn):
7     """
8     Adds a book to the library database.
9     Parameters:
10         title (str): The title of the book.
11         author (str): The author of the book.
12         isbn (str): The unique ISBN number of the book.
13     Returns:
14         str: A message indicating whether the book was added successfully or if it already exists.
15     """
16     if isbn not in library_db:
17         library_db[isbn] = {"title": title, "author": author}
18         return "Book added successfully."
19     else:
20         return "Book already exists."
21
22
23 def remove_book(isbn):
24     """
25     Removes a book from the library database based on its ISBN.
26     Parameters:
27         isbn (str): The unique ISBN number of the book to be removed.
28     Returns:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python Debug Console

- ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54604 -- /Users/ishamadeeha/Documents/AI-assisted\ code/library.py
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 main.py
- Book added successfully.
- Book added successfully.
- {'title': 'Python Basics', 'author': 'John Doe'}
- Book removed successfully.
- Book not found.
- ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pydoc -w library
- wrote library.html
- ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 50, Col 33 Spaces: 4 UTF-8 LF {} Python Python 3.9.6 Go Live



[index](#)
library `/Users/ishamadeeha/Documents/AI-assisted code/library.py`

library.py

Functions

add_book(title, author, isbn)
Adds a book to the library database.
Parameters:
title (str): The title of the book.
author (str): The author of the book.
isbn (str): The unique ISBN number of the book.
Returns:
str: A message indicating whether the book was added successfully or if it already exists.

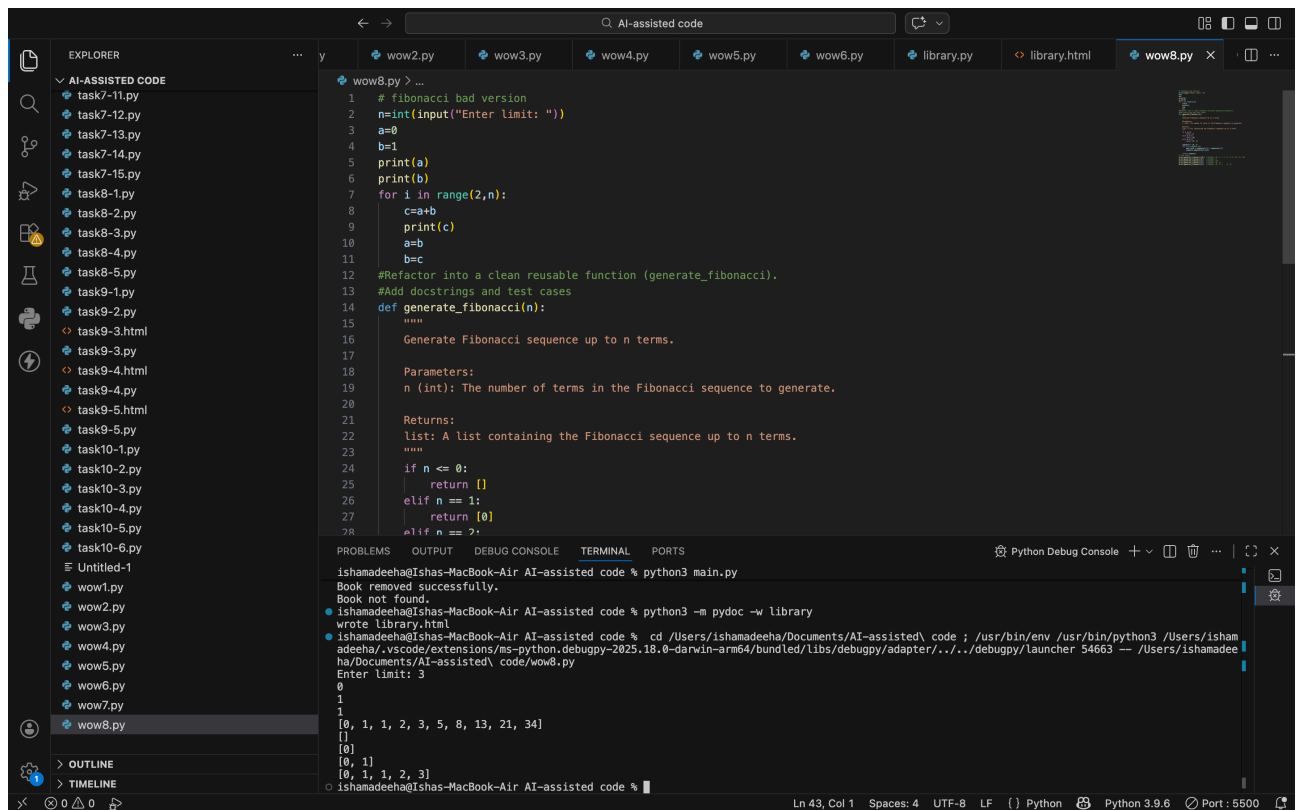
remove_book(isbn)
Removes a book from the library database based on its ISBN.
Parameters:
isbn (str): The unique ISBN number of the book to be removed.
Returns:
str: A message indicating whether the book was removed successfully or if it was not found.

search_book(isbn)
Searches for a book in the library database based on its ISBN.
Parameters:
isbn (str): The unique ISBN number of the book to be searched.
Returns:
dict or str: A dictionary containing the book's title and author if found, otherwise a message indicating that the book was not found.

Data

library_db = {}

TASK-08:



TASK-09:

```
8 fb=0
9 for i in range(2,b):
10     if b%i==0:
11         fb=1
12 if fa==0 and fb==0 and abs(a-b)==2:
13     print("Twin Primes")
14 else:
15     print("Not Twin Primes")
16 #Refactor into is_prime(n) and is_twin_prime(p1, p2).
17 #Add docstrings and optimize
18 #Generate a list of twin primes in a given range
19 def is_prime(n):
20     """
21     Check if a number is prime.
22
23     Parameters:
24     n (int): The number to check for primality.
25
26     Returns:
27     bool: True if n is prime, False otherwise.
28     """
29     if n <= 1:
30         return False
31     for i in range(2, int(n**0.5) + 1):
32         if n % i == 0:
33             return False
34     return True
35 def is_twin_prime(p1, p2):
```

Python Debug Console

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % python3 -m pydoc -w library
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54663 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow9.py
Enter limit: 3
0
1
1
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
[1]
[0]
[0, 1]
[0, 1, 1, 2, 3]
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54752 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow9.py
Twin Primes
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-10:

```
1 def get_zodiac(year):
2     """
3     Determine the Chinese Zodiac sign for a given year.
4
5     Parameters:
6     year (int): The year for which the zodiac sign is required.
7
8     Returns:
9     str: The corresponding Chinese Zodiac animal.
10    """
11    zodiac_cycle = [
12        "Monkey", # 0
13        "Rooster", # 1
14        "Dog", # 2
15        "Pig", # 3
16        "Rat", # 4
17        "Ox", # 5
18        "Tiger", # 6
19        "Rabbit", # 7
20        "Dragon", # 8
21        "Snake", # 9
22        "Horse", # 10
23        "Goat" # 11
24    ]
25
26    return zodiac_cycle[year % 12]
```

Python Debug Console

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54663 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow9.py
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
[1]
[0]
[0, 1]
[0, 1, 1, 2, 3]
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54752 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow9.py
Twin Primes
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54800 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow10.py
Dog
Enter a year: 2006
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-11:

The screenshot shows the VS Code editor with the Explorer sidebar on the left displaying a list of files under 'AI-ASSISTED CODE'. The main editor window shows the code for 'wow11.py'. The code implements a Harshad Number Checker (Unstructured Version) and a Refactored Version using Functions. The Harshad Number Checker takes an input number and checks if it is a Harshad number by dividing it by the sum of its digits. The Refactored Version uses a reusable function 'is_harshad(number)' and a helper function 'sum_of_digits(n)'. The terminal output shows the execution of the code, including the input '88' and the output '88 is not a Harshad number'.

```
1 # Harshad Number Checker (Unstructured Version)
2 num = int(input("Enter a number: "))
3 temp = num
4 sum_digits = 0
5 while temp > 0:
6     digit = temp % 10
7     sum_digits = sum_digits + digit
8     temp = temp // 10
9 if sum_digits != 0:
10     if num % sum_digits == 0:
11         print("True")
12     else:
13         print("False")
14 else:
15     print("False")
16 # Refactored Version using Functions Create a reusable function: is_harshad(number) The function must:Accept an integer pa
17 #!/usr/bin/env python3
18 def sum_of_digits(n):
19     """Calculate the sum of digits of a number."""
20     total = 0
21     while n > 0:
22         digit = n % 10
23         total += digit
24         n //= 10
25     return total
26 def is_harshad(number):
27     """Check if a number is a Harshad number."""
28     if number <= 0:
29         return False
30     return number % sum_of_digits(number) == 0
```

Terminal Output:

```
ha/Documents/AI-assisted\ code\wow11.py
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54752 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow11.py
Twin Primes
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54800 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow10.py
Enter a year: 2006
Dog
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55010 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow10.py
Enter a number: 153
True
Enter a number: 88
88 is not a Harshad number.
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-12:

The screenshot shows the VS Code editor with the Explorer sidebar on the left displaying a list of files under 'AI-ASSISTED CODE'. The main editor window shows the code for 'wow12.py'. The code implements a Trailing Zeros Counter (Unstructured Version) and a Refactored Version using Functions. The Trailing Zeros Counter takes an input number and counts the number of trailing zeros in its factorial. The Refactored Version uses a reusable function 'count_trailing_zeros(n)' and a helper function 'power_of_5(n)'. The terminal output shows the execution of the code, including the input '900' and the output 'Trailing zeros: 27222'.

```
1 # Factorial Trailing Zeros (Unstructured Version)
2 n = int(input("Enter a number: "))
3 fact = 1
4 i = 1
5 while i <= n:
6     fact = fact * i
7     i = i + 1
8 count = 0
9 while fact % 10 == 0:
10     count = count + 1
11     fact = fact // 10
12 print("Trailing zeros:", count)
13 # Refactored Version using Functions 1. Create a reusable function: count_trailing_zeros(n)
14 # The function must:
15 # Accept a non-negative integer n.
16 #!/usr/bin/env python3
17 def count_trailing_zeros(n):
18     """Return the number of trailing zeros in n!."""
19     if n < 0:
20         raise ValueError("Input must be a non-negative integer.")
21     count = 0
22     power_of_5 = 5
23     while n >= power_of_5:
24         count += n // power_of_5
25         power_of_5 *= 5
26     return count
27 def main():
28     # Get user input
```

Terminal Output:

```
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 54800 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow10.py
Enter a number: 153
True
Enter a number: 88
88 is not a Harshad number.
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55010 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow11.py
Enter a number: 153
True
Enter a number: 88
88 is not a Harshad number.
ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 55035 -- /Users/ishamadeeha/Documents/AI-assisted\ code\wow12.py
Enter a number: 900
900
700
Trailing zeros: 27222
Enter a non-negative integer: The number of trailing zeros in 900! is: 224
```

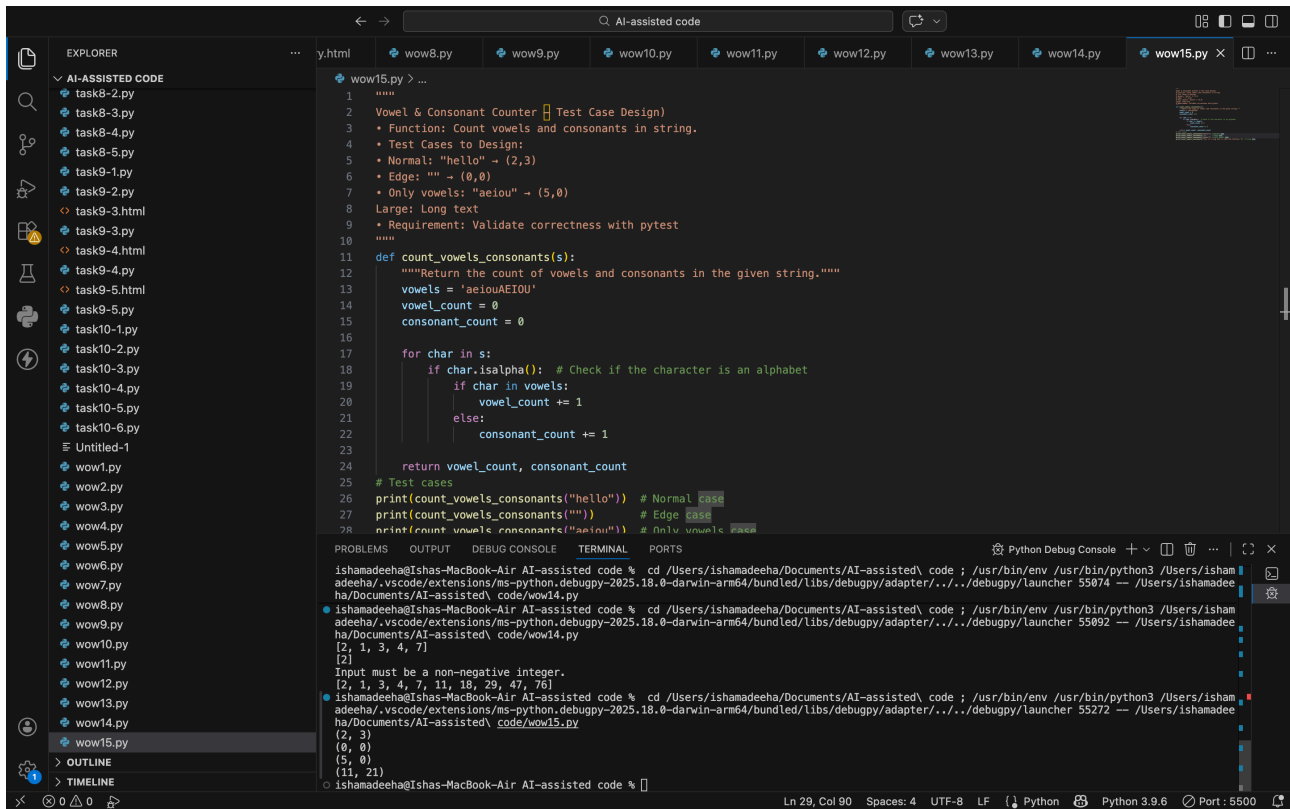

TASK-13:

```
1 #!/usr/bin/env python3
2 """
3 Example
4 Input: 6
5 Steps:
6 6 (even -> 6/2 = 3)
7 3 (odd -> 3*3+1 = 10)
8 10 (even -> 10/2 = 5)
9 5 (odd -> 3*5+1 = 16)
10 16 (even -> 16/2 = 8)
11 8 (even -> 8/2 = 4)
12 4 (even -> 4/2 = 2)
13 2 (even -> 2/2 = 1)
14 Output:
15 [6, 3, 10, 5, 16, 8, 4, 2, 1]
16
17 Function: Generate Collatz sequence until reaching 1. Test Cases to Design: Normal: 6 -> [6,3,10,5,16,8,4,2,1]Edge: 1 -> [1]N
18 """
19 def collatz_sequence(n):
20     """Generate the Collatz sequence for a given integer n."""
21     if n <= 0:
22         raise ValueError("Input must be a positive integer.")
23
24     sequence = []
25     while n != 1:
26         sequence.append(n)
27         if n % 2 == 0:
28             n = n // 2
29         else:
30             n = 3 * n + 1
31
32     sequence.append(1)
33     return sequence
34
35 # Test Cases
36 print(collatz_sequence(6))
37 print(collatz_sequence(1))
38 print(collatz_sequence(10))
39 print(collatz_sequence(16))
40 print(collatz_sequence(8))
41 print(collatz_sequence(4))
42 print(collatz_sequence(2))
43 print(collatz_sequence(1))
44
45 # Edge Case
46 print(collatz_sequence(1))
47
48 # Large Number
49 print(collatz_sequence(900))
50
51 # Trailing zeros
52 print(collatz_sequence(900))
53
54 # Non-negative integer
55 print(collatz_sequence(0))
56
57 # Negative integer
58 print(collatz_sequence(-1))
59
60 # Large integer
61 print(collatz_sequence(76))
62
63 # Requirement: Validate correctness with pytest
64
65 """
66 Trailing zeros: 27222
67 Enter a non-negative integer: The number of trailing zeros in 900! is: 224
68 ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapters/.../debugpy/launcher 55035 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow13.py
69 700
70 Trailing zeros: 27222
71 Enter a non-negative integer: The number of trailing zeros in 900! is: 224
72 ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapters/.../debugpy/launcher 55035 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow13.py
73 Enter a positive integer: 55
74 Collatz Sequence: [55, 166, 83, 250, 125, 376, 188, 94, 47, 142, 71, 214, 107, 322, 161, 484, 242, 121, 364, 182, 91, 274, 137, 412, 206, 103, 3, 10, 155, 466, 233, 700, 350, 175, 526, 263, 790, 395, 1186, 593, 1780, 890, 445, 1336, 668, 334, 167, 502, 251, 754, 377, 1132, 566, 283, 850, 4, 25, 1276, 638, 319, 958, 479, 1438, 719, 2158, 1079, 3238, 1619, 4858, 2429, 7288, 3644, 1822, 911, 2734, 1367, 4102, 2051, 6154, 3077, 9232, 46, 16, 2380, 1154, 577, 1732, 866, 433, 1300, 650, 325, 976, 488, 244, 122, 61, 184, 92, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1]
75 ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-14:

```
1 """
2 (Lucas Number Sequence) Test Case Design
3 * Function: Generate Lucas sequence up to n terms.
4 (Starts with 2,1, then Fn = Fn-1 + Fn-2)
5 * Test Cases to Design:
6 * Normal: 5 -> [2, 1, 3, 4, 7]
7 * Edge: 1 -> [2]
8 * Negative: -5 - Error
9 * Large: 10 (last element = 76).
10 * Requirement: Validate correctness with pytest
11 """
12 def lucas(n):
13     if n < 0:
14         raise ValueError("Input must be a non-negative integer.")
15     elif n == 0:
16         return []
17     elif n == 1:
18         return [2]
19     elif n == 2:
20         return [2, 1]
21
22     sequence = [2, 1]
23     for i in range(2, n):
24         next_value = sequence[i-1] + sequence[i-2]
25         sequence.append(next_value)
26
27     return sequence
28
29 # Test Cases
30 print(lucas(5))
31 print(lucas(1))
32 print(lucas(2))
33 print(lucas(3))
34 print(lucas(4))
35 print(lucas(5))
36 print(lucas(6))
37 print(lucas(7))
38 print(lucas(8))
39 print(lucas(9))
40 print(lucas(10))
41 print(lucas(11))
42 print(lucas(12))
43 print(lucas(13))
44 print(lucas(14))
45 print(lucas(15))
46 print(lucas(16))
47 print(lucas(17))
48 print(lucas(18))
49 print(lucas(19))
50 print(lucas(20))
51 print(lucas(21))
52 print(lucas(22))
53 print(lucas(23))
54 print(lucas(24))
55 print(lucas(25))
56 print(lucas(26))
57 print(lucas(27))
58 print(lucas(28))
59 print(lucas(29))
60 print(lucas(30))
61 print(lucas(31))
62 print(lucas(32))
63 print(lucas(33))
64 print(lucas(34))
65 print(lucas(35))
66 print(lucas(36))
67 print(lucas(37))
68 print(lucas(38))
69 print(lucas(39))
70 print(lucas(40))
71 print(lucas(41))
72 print(lucas(42))
73 print(lucas(43))
74 print(lucas(44))
75 print(lucas(45))
76 print(lucas(46))
77 print(lucas(47))
78 print(lucas(48))
79 print(lucas(49))
80 print(lucas(50))
81 print(lucas(51))
82 print(lucas(52))
83 print(lucas(53))
84 print(lucas(54))
85 print(lucas(55))
86 print(lucas(56))
87 print(lucas(57))
88 print(lucas(58))
89 print(lucas(59))
90 print(lucas(60))
91 print(lucas(61))
92 print(lucas(62))
93 print(lucas(63))
94 print(lucas(64))
95 print(lucas(65))
96 print(lucas(66))
97 print(lucas(67))
98 print(lucas(68))
99 print(lucas(69))
100 print(lucas(70))
101 print(lucas(71))
102 print(lucas(72))
103 print(lucas(73))
104 print(lucas(74))
105 print(lucas(75))
106 print(lucas(76))
107 print(lucas(77))
108 print(lucas(78))
109 print(lucas(79))
110 print(lucas(80))
111 print(lucas(81))
112 print(lucas(82))
113 print(lucas(83))
114 print(lucas(84))
115 print(lucas(85))
116 print(lucas(86))
117 print(lucas(87))
118 print(lucas(88))
119 print(lucas(89))
120 print(lucas(90))
121 print(lucas(91))
122 print(lucas(92))
123 print(lucas(93))
124 print(lucas(94))
125 print(lucas(95))
126 print(lucas(96))
127 print(lucas(97))
128 print(lucas(98))
129 print(lucas(99))
130 print(lucas(100))
131
132 # Edge Case
133 print(lucas(1))
134
135 # Large Number
136 print(lucas(76))
137
138 # Requirement: Validate correctness with pytest
139
140 """
141 Input must be a non-negative integer.
142 [2, 1, 3, 4, 7, 11, 18, 29, 47, 76]
143 ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/Libs/debugpy/adapters/.../debugpy/launcher 55092 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow14.py
144 [2, 1, 3, 4, 7]
145 Input must be a non-negative integer.
146 [2, 1, 3, 4, 7, 11, 18, 29, 47, 76]
147 ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```


TASK-15:



The image shows a VS Code editor interface with a file explorer on the left and a terminal at the bottom. The file explorer lists several files under 'AI-ASSISTED CODE', including task8-2.py through task10-6.py, and wow1.py through wow15.py. The active file is wow15.py, which contains a Python script for counting vowels and consonants in a string. The script includes a function `count_vowels_consonants(s)` that returns a tuple of vowel and consonant counts. The terminal window shows the execution of the script for various test cases, including 'hello', an empty string, and 'aeiou', all of which pass the tests.

```
1 """
2 Vowel & Consonant Counter (Test Case Design)
3 • Function: Count vowels and consonants in string.
4 • Test Cases to Design:
5   • Normal: "hello" -> (2,3)
6   • Edge: "" -> (0,0)
7   • Only vowels: "aeiou" -> (5,0)
8   • Large: Long text
9   • Requirement: Validate correctness with pytest
10 """
11 def count_vowels_consonants(s):
12     """Return the count of vowels and consonants in the given string."""
13     vowels = 'aeiouAEIOU'
14     vowel_count = 0
15     consonant_count = 0
16
17     for char in s:
18         if char.isalpha(): # Check if the character is an alphabet
19             if char in vowels:
20                 vowel_count += 1
21             else:
22                 consonant_count += 1
23
24     return vowel_count, consonant_count
25
26 # Test cases
27 print(count_vowels_consonants("hello")) # Normal case
28 print(count_vowels_consonants("")) # Edge case
29 print(count_vowels_consonants("aeiou")) # Only vowels case
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 55074 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow14.py

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 55092 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow14.py

[2, 1, 3, 4, 7]

[2]

Input must be a non-negative integer.

[2, 1, 3, 4, 7, 11, 19, 29, 47, 76]

ishamadeeha@Ishas-MacBook-Air AI-assisted code % cd /Users/ishamadeeha/Documents/AI-assisted\ code ; /usr/bin/env /usr/bin/python3 /Users/ishamadeeha/.vscode/extensions/ms-python.debugpy-2025.18.0-darwin-arm64/bundled/libs/debugpy/adapter/.../debugpy/launcher 55272 -- /Users/ishamadeeha/Documents/AI-assisted\ code/wow15.py

(2, 3)

(0, 0)

(5, 0)

(11, 21)

ishamadeeha@Ishas-MacBook-Air AI-assisted code %

Ln 29, Col 90 Spaces: 4 UTF-8 LF Python Python 3.9.6 Port: 5500